

# Report

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## 1.

We will use the fact that in optimal policy,  $\frac{\partial \mathcal{L}}{\partial t} = 0$ .

$$0 = \frac{\partial \mathcal{L}}{\partial t} \tag{1}$$

$$= - \int_i G' \left( U'_{ic} \left( \frac{\partial z_i}{\partial t} - z_i - t \frac{\partial z_i}{\partial t} \right) + U'_{iz} \frac{\partial z_i}{\partial t} \right) dv(i) + \lambda \int_i \left( z_i + t \frac{\partial z_i}{\partial t} \right) dv(i) \tag{2}$$

$$= - \int_i z_i U'_{ic} G' dv(i) + \lambda \int_i \left( z_i + t \frac{\partial z_i}{\partial t} \right) dv(i) \tag{3}$$

Rearranging and using  $g_i$  and  $\varepsilon_i$ , we get:

$$\int_i g_i z_i dv(i) = \int_i \left( z_i - \varepsilon_i \frac{t}{1-t} z_i \right) dv(i).$$

After using the fact that  $\varepsilon_i = \varepsilon_i^c = \eta_i$  and  $g_i + \frac{t}{1-t} \eta_i = \tilde{g}_i$ , we can simplify the equation to:

$$\int_i \tilde{g}_i z_i dv(i) = \int_i \left( z_i - \varepsilon_i^c \frac{t}{1-t} z_i \right) dv(i) \Rightarrow - \int_i \varepsilon_i^c \frac{t}{1-t} z_i dv(i) = \int_i \tilde{g}_i z_i - z_i dv(i).$$

Since expectation of  $\tilde{g}_i$  is 1, we can denote the right hand side of the equation as  $\text{Cov}(\tilde{g}_i, z_i)$ . Then assuming elasticity is same for everyone, we can get the equation in the question 7.

## Part 2: Numerical application

```
# Load libraries
library(tidyverse)
library(gt)

# Given parameters
t <- 0.3 # Current tax rate
b <- 5000 # Lump-sum benefit
```

```
implied_ability <- map(c(20000, 50000, 200000), ~ (.x / (1-t)^0.3)^(1/1.3))
  ↪ |>
  bind_cols() |>
  rename_with(~ c("low", "middle", "high")) |> print()
```

```
# A tibble: 1 x 3
  low middle high
<dbl> <dbl> <dbl>
1 2209. 4470. 12986.
```